

DD: OSR-RMA-01-FLOW-SWAP-HFC — HFC field-pickup equipment swap

Developer implementation guide: DD_OSR-RMA-01-FLOW-SWAP-HFC-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the operator BPMN flow DD_OSR-RMA-01-FLOW-SWAP-HFC-v1.0.md.

Verdict: ■ New | **Wave:** 2 | **Group:** OSR (Operator Stock Registry) | **Version:** v1.0 | **Module:** HFC Equipment Swap Flow
Satellite of: OSR-RMA-01 Equipment Swap & RMA Per-flow satellite for the HFC field-pickup equipment swap (swap_kind=HFC_FIELD). Owns the BPMN EquipmentSwap_HFC_WIK.bpmn + per-flow REST endpoint + per-flow Kafka payload shape + flow-specific Drools rules + worked end-to-end example. Reuses framework workers, messages, user tasks, history schema, and FOUNDATION_AUTH groups. v1.0 scope: KE WIK only. WUG/WTZ/YASSN variants via separate BPMN files when workshops land.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/08_osr_equipment/DD_OSR-RMA-01-FLOW-SWAP-HFC-v1.0.md
Version	v1.0
DD type	operator BPMN flow
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- GET /api/swap-requests/swap_01HZ_NJERI_MODEM
- GET /api/swap-requests/{id}
- POST /api/billable-events
- POST /api/contractor-slot-commitments
- POST /api/equipment-instances/inst_01HZ_CM_NJERI_NEW/lifecycle-events
- POST /api/equipment-instances/inst_01HZ_CM_NJERI_OLD/lifecycle-events

- POST /api/stock-movements
- POST /api/swap-requests/hfc
- POST /api/work-orders
- POST /broadhub-adapter/v1/serial/rebind
- POST /broadhub-adapter/v1/serial/unbind

4.2 Tables

- No CREATE TABLE block is explicitly declared in this DD.

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- bil01-charge-pre-authorize
- cc-agent-wik-007
- contractor-slot-reserve
- contractor-stock-route
- customer-appointment-accepted
- homepass-network-path-update
- instance-state-transition
- nrb-cmts-01
- payment-confirmed
- serial-rebind-broadhub-stub
- serial-unbind-broadhub-stub
- swap-request-history-append
- tech-confirm-defect
- wo-finalized

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- WorkOrderFinalized

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ New | **Wave:** 2 | **Group:** OSR (Operator Stock Registry) | **Version:** v1.0 | **Module:** HFC Equipment Swap Flow

Satellite of: OSR-RMA-01 Equipment Swap & RMA

Per-flow satellite for the HFC field-pickup equipment swap (swap_kind=HFC_FIELD). Owns the BPMN EquipmentSwap_HFC_WIK.bpmn + per-flow REST endpoint + per-flow Kafka payload shape + flow-specific Drools rules + worked end-to-end example. Reuses framework workers, messages, user tasks, history schema, and FOUNDATION_AUTH groups.

v1.0 scope: KE WIK only. WUG/WTZ/YASSN variants via separate BPMN files when workshops land.

1. What this flow is

When a customer's HFC cable modem (or hybrid chassis with MTA, or STB, or smart card) fails — or NOC proactively detects an imminent failure, or the customer requests an in-tech upgrade — a technician must visit the premises with a replacement unit, retrieve the defective unit, and rebind the new device's CM-MAC at the SAME CMTS slot/port as the old one (UC-PR4: HFC modem authorization is bound to slot/port, moving the modem to a different port breaks authorization).

This flow is **always** field-visit (no remote-swap path for HFC — DOCSIS modems can't be unprovisioned remotely without the unit being physically present for the swap), and it's **always** contractor-routed (the tech who recovers the old modem becomes the recovery contractor; the defective unit routes back to that contractor's van, not main warehouse — the Wananchi 40+ stale-equipment fix).

When the source instance is a hybrid chassis (HYB equipment_kind, modeled per workshop 2026-04-29 as CM-MAC + MTA-MAC + optional CPMAC for static-IP packages), the Broadhub double-config path applies: Broadhub + Clearcable NOMS are both reconfigured by the serial-rebind-broadhub-stub worker.

What this flow does NOT do

- **Does not handle GPON ONT swaps.** Use swap_kind=GPON_REMOTE or GPON_FIELD (separate FLOW-SWAP-GPON satellite). HFC and GPON have different provisioning adapters and different OLT-side semantics.
- **Does not handle equipment termination retrieval** — that's EQP_TERMINATION (separate FLOW-EQP satellite).
- **Does not orchestrate the WO itself.** WO creation calls WO-01; WO lifecycle (claim, dispatch, finalize) is WO-01's concern. This flow reacts to wo-finalized message catch.
- **Does not own the contractor-stock-routing logic.** Uses framework worker contractor-stock-route which encapsulates the Wananchi-fix routing decision (defective unit → recovering contractor's van).
- **Does not handle HomePass.** HFC equipment-to-HomePass binding is NOT modeled in HomePass-CFG network_path today (HomePass.network_path is GPON-centric, terminal ONT node). HFC swaps therefore do not call homepass-network-path-update.

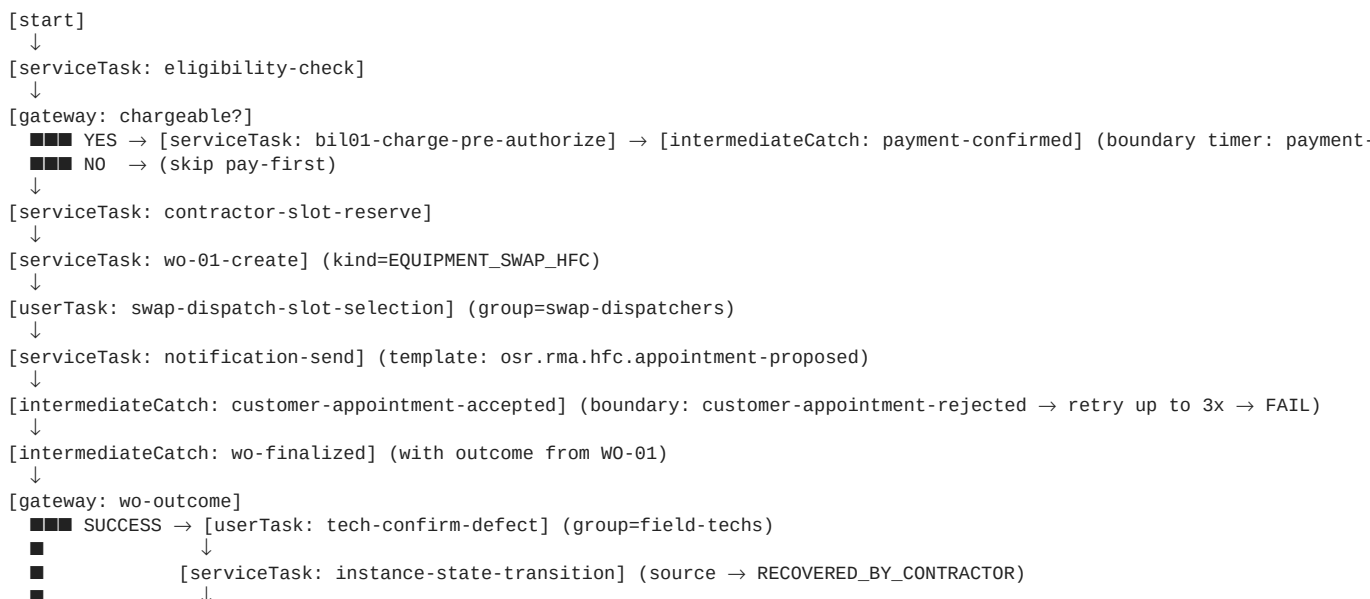
Glossary additions (specific to this flow)

Term	Meaning
CMTS slot/port	The physical CMTS (Cable Modem Termination System) chassis card slot and port number to which the HFC modem is bound for authorization. Format {cmts:'nrb-cmts-01', slot:3, port:12}.
UC-PR4 (slot/port preservation)	Workshop 2026-04-30 rule: HFC modem authorization MUST stay at the same slot/port across a swap. Moving the new modem to a different slot/port breaks Broadhub authorization. The rebind call MUST carry the source's slot/port verbatim.
Double-config (static-IP HFC)	For static-IP HFC packages, two provisioning systems must be reconfigured: Broadhub (subscriber + CM-MAC binding) AND Clearcable NOMS (the CPMAC public-IP binding). Single rebind call exposes both; adapter handles the dual write.
Hybrid chassis	An HFC equipment chassis carrying both a cable modem (CM-MAC) and an MTA voice port (MTA-MAC). Per workshop 2026-04-29, MTA-MAC = CM-MAC + 2 (computed). Some hybrid chassis also carry a CPMAC for static-IP carriage.

2. BPMN — EquipmentSwap_HFC_WIK.bpmn

2.1 Process shape

The BPMN starts on SwapRequestCreated for swap_kind=HFC_FIELD and drives the swap-request through its states until terminal.



```

■      [serviceTask: contractor-stock-route]
■      ↓
■      [serviceTask: instance-state-transition] (target → IN_FIELD_ACTIVE)
■      ↓
■      [serviceTask: serial-unbind-broadhub-stub] (old CM-MAC at slot/port)
■      ↓
■      [serviceTask: serial-rebind-broadhub-stub] (new CM-MAC at SAME slot/port – UC-PR4)
■      ↓
■      [gateway: rebind-ok?]
■          ■■■ YES → [end: COMPLETED]
■          ■■■ NO → [retry x3] → [end: FAILED]
■      ■■■ CUSTOMER_REFUSED_APPOINTMENT → [end: FAILED]
■      ■■■ (other) → [end: FAILED]

```

2.2 BPMN XML — key snippets

Chargeable gateway + pay-first wait:

```

<bpmn:exclusiveGateway id="Gateway_Chargeable" name="Chargeable?" />
<bpmn:sequenceFlow id="Flow_ChargeableYes" sourceRef="Gateway_Chargeable" targetRef="Task_PreAuthCharge">
  <bpmn:conditionExpression xsi:type="bpmn:tFormalExpression">${chargeable == true}</bpmn:conditionExpression>
</bpmn:sequenceFlow>
<bpmn:sequenceFlow id="Flow_ChargeableNo" sourceRef="Gateway_Chargeable" targetRef="Task_ContractorSlotReserve">
  <bpmn:conditionExpression xsi:type="bpmn:tFormalExpression">${chargeable == false}</bpmn:conditionExpression>
</bpmn:sequenceFlow>

<bpmn:serviceTask id="Task_PreAuthCharge" name="Pre-authorize BIL-01 charge"
  camunda:type="external" camunda:topic="bil01-charge-pre-authorize" />

<bpmn:intermediateCatchEvent id="Catch_PaymentConfirmed" name="Payment confirmed">
  <bpmn:messageEventDefinition messageRef="msg_payment_confirmed" />
</bpmn:intermediateCatchEvent>

<bpmn:boundaryEvent id="Timer_PaymentTimeout" attachedToRef="Catch_PaymentConfirmed" cancelActivity="true">
  <bpmn:timerEventDefinition>
    <bpmn:timeDuration>P7D</bpmn:timeDuration>
  </bpmn:timerEventDefinition>
</bpmn:boundaryEvent>

```

Rebind with UC-PR4 slot/port preservation:

```

<bpmn:serviceTask id="Task_RebindBroadhub" name="Rebind new CM-MAC at SAME slot/port"
  camunda:type="external" camunda:topic="serial-rebind-broadhub-stub">
  <bpmn:extensionElements>
    <camunda:inputOutput>
      <camunda:inputParameter name="instanceId">${targetInstanceId}</camunda:inputParameter>
      <camunda:inputParameter name="cmMac">${targetCmMac}</camunda:inputParameter>
      <camunda:inputParameter name="mtaMac">${targetMtaMac}</camunda:inputParameter>
      <camunda:inputParameter name="cpMac">${targetCpMac}</camunda:inputParameter>
      <camunda:inputParameter name="slotPort">${flowPayload.slotPort}</camunda:inputParameter>
      <camunda:inputParameter name="doubleConfig">${flowPayload.doubleConfig}</camunda:inputParameter>
    </camunda:inputOutput>
  </bpmn:extensionElements>
</bpmn:serviceTask>

```

2.3 Per-operator BPMN composition

KE WIK ships EquipmentSwap_HFC_WIK.bpmn (this satellite). WUG/WTZ/YASSN: WTZ doesn't have HFC infrastructure (GPON-only Tanzania per workshop sources); YASSN's HFC stance TBD pending Senegal workshop on swap workflow. Variants land as EquipmentSwap_HFC_<OPERATOR>.bpmn files; framework workers are unchanged.

3. Flow_payload shape (HFC_FIELD)

This flow's contribution to equipment_swap_request.flow_payload JSONB:

```

{
  "cmts":          "nrb-cmts-01",
  "slot":          3,
  "port":          12,
  "authzBound":    true,
  "doubleConfig":  false,
  "sourceCmMac":   "00:11:22:33:44:55",
  "sourceMtaMac":  "00:11:22:33:44:57",
  "sourceCpMac":   null,
  "targetCmMac":   "00:AA:BB:CC:DD:EE",
  "targetMtaMac":  "00:AA:BB:CC:DD:F0",
  "targetCpMac":   null,
  "isHybridChassis": true,

```

```

    "isStaticIp":    false
  }

```

Field meanings:

Field	Sourced from	Purpose
cmts, slot, port	Source instance network_slot_metadata	UC-PR4 preservation target — rebind call MUST use these exact values
authzBound	Source instance network_slot_metadata.authz_bound	Operational flag: confirms the slot/port currently has authorization (sanity check at eligibility)
doubleConfig	Eligibility check: derived from subscription package metadata	TRUE iff customer has a static-IP HFC package; drives Clearcable NOMS dual-write
sourceCmMac/MtaMac/CpMac	Source instance	Identifies what to unbind
targetCmMac/MtaMac/CpMac	Target instance	Identifies what to bind at the preserved slot/port
isHybridChassis	PLM-CFG-06 SKU equipment_kind=HYB	Drives MTA-MAC presence requirement
isStaticIp	Subscription package metadata	Confirms Clearcable NOMS path needed

4. Rules (HFC-specific)

Rule	Description
R-HFC-1	Source equipment_kind MUST be CABLE_MODEM, MTA, STB, SMART_CARD, or HYB. Eligibility worker rejects with failureCode=NOT_HFC_FAMILY otherwise.
R-HFC-2	Source network_slot_metadata.cmts, slot, port MUST be non-null. Eligibility worker rejects with failureCode=MISSING_SLOT_PORT otherwise — slot/port preservation impossible without source values.
R-HFC-3	Target instance MUST have an SKU compatible with the source's equipment_kind (e.g., source CM → target CM; source HYB → target HYB; cross-kind rejected). Future enhancement: explicit upgrade-compatibility matrix per operator in PLM-CFG-06.
R-HFC-4	Hybrid chassis target MUST carry CM-MAC + MTA-MAC + (CPMAC if source had CPMAC). Eligibility worker validates; failureCode=HYB_MAC_INCOMPLETE.
R-HFC-5	serial-rebind-broadhub-stub worker MUST receive slotPort matching flow_payload.{cmts, slot, port} from source. Worker-side assertion; mismatch results in immediate FAILED transition with reason=SLOT_PORT_MISMATCH_AT_REBIND (R-OSR-RMA-8 enforces).
R-HFC-6	Static-IP packages require doubleConfig=true in flow_payload; adapter receives the dual-write signal. Eligibility worker derives this from subscription metadata; if metadata is missing, conservative default is FALSE (single-write).
R-HFC-7	Rebind retry: up to 3 attempts before transitioning to FAILED. Each retry triggers a new lifecycle event row (PROVISIONING_RETRY_<n>). Final failure rolls back source instance to van (IN_CONTRACTOR_STOCK) and target instance to van (IN_CONTRACTOR_STOCK); no customer-side change.
R-HFC-8	If source CMTS slot/port differs from any of the operator's known active CMTS chassis (operator-config cmts_registry), eligibility fails with failureCode=CMTS_UNKNOWN. Prevents swap creation against stale or migrated CMTS references.

5. REST API

```

POST /api/swap-requests/hfc HTTP/1.1
Authorization: Bearer <noc-jwt>
Content-Type: application/json

```

```

{
  "operatorCode":    "WIK",
  "accountId":       "acct_01HZ_NJERI",
  "subscriptionId":  "sub_01HZ_NJERI_TRIPLEPLAY",
  "sourceInstanceId": "inst_01HZ_CM_NJERI_OLD",
  "targetInstanceId": "inst_01HZ_CM_NJERI_NEW",
  "reasonCode":      "CUSTOMER_REPORTED_DEFECT",
  "initiatedByUserId": "cc-agent-wik-007",
  "notes":           "Customer reports modem reboots every 15 min for past 3 days"
}

```

```

→ 201 Created
{
  "swapRequestId": "swap_01HZ_NJERI_MODEM",
  "operatorCode": "WIK",
  "swapKind": "HFC_FIELD",
  "currentState": "ELIGIBILITY_CHECK",
  "accountId": "acct_01HZ_NJERI",
  "subscriptionId": "sub_01HZ_NJERI_TRIPLEPLAY",
  "sourceInstanceId": "inst_01HZ_CM_NJERI_OLD",
  "targetInstanceId": "inst_01HZ_CM_NJERI_NEW",
  "reasonCode": "CUSTOMER_REPORTED_DEFECT",
  "initiatedByUserId": "cc-agent-wik-007",
  "initiatedAt": "2026-05-22T08:30:00Z",
  "bpmnProcessInstanceId": "camunda-pi-7f3a2b8c",
  "chargeable": null,
  "createdAt": "2026-05-22T08:30:00Z"
}

```

The response 201 fires the BPMN. Subsequent state transitions visible via framework GET /api/swap-requests/{id}.

After eligibility check:

```

GET /api/swap-requests/swap_01HZ_NJERI_MODEM HTTP/1.1
Authorization: Bearer <cc-agent-jwt>

```

```

→ 200 OK
{
  "swapRequestId": "swap_01HZ_NJERI_MODEM",
  "currentState": "AWAITING_PAYMENT",
  "chargeable": true,
  "chargeCode": "EQUIPMENT_SWAP_OUT_OF_WARRANTY",
  "customerChargeAmount": 3500.00,
  "billableEventId": "bevt_01HZ_NJERI_SWAP_CHARGE",
  "flowPayload": {
    "cmts": "nrb-cmts-01",
    "slot": 3,
    "port": 12,
    "authzBound": true,
    "doubleConfig": false,
    "sourceCmMac": "00:11:22:33:44:55",
    "sourceMtaMac": "00:11:22:33:44:57",
    "targetCmMac": "00:AA:BB:CC:DD:EE",
    "targetMtaMac": "00:AA:BB:CC:DD:F0",
    "isHybridChassis": true,
    "isStaticIp": false
  },
  "updatedAt": "2026-05-22T08:30:42Z"
}

```

6. Kafka events emitted

Topic prefix: `sophix.osr.rma.*` (framework-defined; events populated by this satellite at the corresponding state transitions).

6.1 SwapRequestCreated

```

{
  "eventType": "SwapRequestCreated",
  "aggregateId": "swap_01HZ_NJERI_MODEM",
  "timestamp": "2026-05-22T08:30:00.000Z",
  "payload": {
    "operatorCode": "WIK",
    "swapRequestId": "swap_01HZ_NJERI_MODEM",
    "swapKind": "HFC_FIELD",
    "accountId": "acct_01HZ_NJERI",
    "subscriptionId": "sub_01HZ_NJERI_TRIPLEPLAY",
    "sourceInstanceId": "inst_01HZ_CM_NJERI_OLD",
    "targetInstanceId": "inst_01HZ_CM_NJERI_NEW",
    "reasonCode": "CUSTOMER_REPORTED_DEFECT",
    "initiatedByUserId": "cc-agent-wik-007"
  }
}

```

6.2 SwapRequestAwaitingPayment

```

{
  "eventType": "SwapRequestAwaitingPayment",
  "aggregateId": "swap_01HZ_NJERI_MODEM",

```

```

    "timestamp": "2026-05-22T08:30:42.000Z",
    "payload": {
      "operatorCode": "WIK",
      "swapRequestId": "swap_01HZ_NJERI_MODEM",
      "swapKind": "HFC_FIELD",
      "accountId": "acct_01HZ_NJERI",
      "chargeCode": "EQUIPMENT_SWAP_OUT_OF_WARRANTY",
      "customerChargeAmount": 3500.00,
      "billableEventId": "bevt_01HZ_NJERI_SWAP_CHARGE",
      "paymentDeadline": "2026-05-29T08:30:42.000Z"
    }
  }
}

```

6.3 SwapRequestSourceRecovered

```

{
  "eventType": "SwapRequestSourceRecovered",
  "aggregateId": "swap_01HZ_NJERI_MODEM",
  "timestamp": "2026-05-25T14:22:00.000Z",
  "payload": {
    "operatorCode": "WIK",
    "swapRequestId": "swap_01HZ_NJERI_MODEM",
    "swapKind": "HFC_FIELD",
    "sourceInstanceId": "inst_01HZ_CM_NJERI_OLD",
    "sourceSerial": "HFC-CHASSIS-44556-OLD",
    "recoveryContractorId": "contractor_ke_nairobi_a",
    "recoveredAt": "2026-05-25T14:22:00+03:00"
  }
}

```

6.4 SwapRequestCompleted

```

{
  "eventType": "SwapRequestCompleted",
  "aggregateId": "swap_01HZ_NJERI_MODEM",
  "timestamp": "2026-05-25T15:10:00.000Z",
  "payload": {
    "operatorCode": "WIK",
    "swapRequestId": "swap_01HZ_NJERI_MODEM",
    "swapKind": "HFC_FIELD",
    "accountId": "acct_01HZ_NJERI",
    "subscriptionId": "sub_01HZ_NJERI_TRIPLEPLAY",
    "sourceInstanceId": "inst_01HZ_CM_NJERI_OLD",
    "targetInstanceId": "inst_01HZ_CM_NJERI_NEW",
    "recoveryContractorId": "contractor_ke_nairobi_a",
    "woId": "wo_01HZ_NJERI_SWAP",
    "chargeCode": "EQUIPMENT_SWAP_OUT_OF_WARRANTY",
    "customerChargeAmount": 3500.00,
    "paidAt": "2026-05-22T11:15:00Z",
    "rebindSlotPort": {"cmts": "nrb-cmts-01", "slot": 3, "port": 12},
    "closedAt": "2026-05-25T15:10:00+03:00"
  }
}

```

7. Worked end-to-end example — Njeri's cable modem swap

Customer: Njeri, WIK Triple-Play subscriber (HFC, hybrid chassis modem+MTA, no static IP). Account acct_01HZ_NJERI. Address served by Nairobi CMTS nrb-cmts-01, slot 3, port 12. Old modem inst_01HZ_CM_NJERI_OLD serial HFC-CHASSIS-44556-OLD, CM-MAC 00:11:22:33:44:55, MTA-MAC 00:11:22:33:44:57. Out of vendor warranty (purchased 2023). Recovery contractor (resolved by tech-region lookup): contractor_ke_nairobi_a (Franchise A).

Day 0 — 08:30: Njeri calls customer care: "my internet reboots every 15 minutes for 3 days." CC agent cc-agent-wik-007 confirms a target unit is in Franchise A's van stock (inst_01HZ_CM_NJERI_NEW, fresh from main warehouse last week, CM-MAC 00:AA:BB:CC:DD:EE). CC agent posts:

```

POST /api/swap-requests/hfc
{
  "operatorCode": "WIK", "accountId": "acct_01HZ_NJERI", "subscriptionId": "sub_01HZ_NJERI_TRIPLEPLAY",
  "sourceInstanceId": "inst_01HZ_CM_NJERI_OLD", "targetInstanceId": "inst_01HZ_CM_NJERI_NEW",
  "reasonCode": "CUSTOMER_REPORTED_DEFECT", "initiatedByUserId": "cc-agent-wik-007",
  "notes": "Customer reports modem reboots every 15 min for past 3 days"
}

```

Response 201: swap_01HZ_NJERI_MODEM, state ELIGIBILITY_CHECK. BPMN starts.

Day 0 — 08:30:42: Eligibility worker runs. Reads source instance → CM_MAC + MTA_MAC + slot/port present. Reads PLM-CFG-06 for source SKU → out-of-warranty (purchase date 2023, warranty 2 years, expired). Resolves charge: chargeable=true, chargeCode=EQUIPMENT_SWAP_OUT_OF_WARRANTY, customerChargeAmount=3500.00 (KES). Builds flow_payload. Writes:

```
UPDATE equipment_swap_request SET
  current_state='AWAITING_PAYMENT',
  chargeable=TRUE,
  charge_code='EQUIPMENT_SWAP_OUT_OF_WARRANTY',
  customer_charge_amount=3500.00,
  flow_payload='{ "cmts": "nrb-cmts-01", "slot": 3, "port": 12, "authzBound": true, "doubleConfig": false, "sourceCmMac": "00:11:22:33:44:55" }'
WHERE swap_request_id='swap_01HZ_NJERI_MODEM';
```

Worker bil01-charge-pre-authorize fires:

```
POST /api/billable-events
{"accountId": "acct_01HZ_NJERI", "chargeCode": "EQUIPMENT_SWAP_OUT_OF_WARRANTY",
 "amount": 3500.00, "contextRef": {"kind": "swap_request", "id": "swap_01HZ_NJERI_MODEM"}}
→ 201 {"billableEventId": "bevt_01HZ_NJERI_SWAP_CHARGE", "status": "AWAITING_PAYMENT"}
```

Kafka emits SwapRequestAwaitingPayment. NOT-01 sends SMS:

"Hi Njeri, your modem swap charge is KES 3,500. Pay via M-Pesa to 123456, reference SWAP-NJERI-MODEM. Once paid, we'll book your appointment. Reply STOP to cancel."

BPMN parks at message catch payment-confirmed with a 7-day boundary timer.

Day 0 — 11:15: Njeri pays KES 3,500 via M-Pesa. Payment gateway → BIL-01 → emits BillableEventPaid Kafka event with billable_event_id=bevt_01HZ_NJERI_SWAP_CHARGE. BPMN message catch fires. Worker swap-request-history-append writes history row. State advances to AWAITING_SLOT. payment_received_at=2026-05-22T11:15:00Z.

Day 0 — 11:16: Worker contractor-slot-reserve calls EM-02:

```
POST /api/contractor-slot-commitments
{"operatorCode": "WIK", "accountId": "acct_01HZ_NJERI", "techRegionId": "tregion_ke_nrb_central",
 "serviceScope": "HFC", "requiredSkills": ["HFC_MODEM_SWAP"],
 "windowStart": "2026-05-23T08:00:00+03:00", "windowEnd": "2026-05-25T18:00:00+03:00"}
→ 201 {"slotCommitmentId": "slot_01HZ_NJERI_SWAP", "contractorId": "contractor_ke_nairobi_a",
 "scheduledFor": "2026-05-25T14:00:00+03:00"}
```

Swap-request updated: recovery_contractor_id='contractor_ke_nairobi_a', slot_commitment_id='slot_01HZ_NJERI_SWAP', state SLOT_RESERVED.

Day 0 — 11:17: Worker calls WO-01:

```
POST /api/work-orders
{"operatorCode": "WIK", "accountId": "acct_01HZ_NJERI", "woKind": "EQUIPMENT_SWAP_HFC",
 "originRef": {"kind": "swap_request", "id": "swap_01HZ_NJERI_MODEM"},
 "contractorId": "contractor_ke_nairobi_a", "scheduledFor": "2026-05-25T14:00:00+03:00",
 "requirements": ["EQUIPMENT_BINDING:source_serial,target_serial", "SLOT_PORT_PRESERVATION:cmts=nrb-cmts-01,slot=3,port=12"]}
→ 201 {"woId": "wo_01HZ_NJERI_SWAP", "status": "DISPATCHED"}
```

State WO_CREATED. WO-01 takes over scheduling/dispatch.

Day 0 — 11:18: NOT-01 sends customer SMS: *"Njeri, your modem swap is scheduled for Mon 25 May, 14:00–18:00. Reply YES to confirm or NO to reschedule."* Njeri replies YES → message catch customer-appointment-accepted fires.

Day 3 — 14:05: Franchise A tech tech_franchise_a_03 arrives at Njeri's premises with the new modem. Opens his mobile app, sees WO wo_01HZ_NJERI_SWAP. Confirms defect (modem indeed unresponsive). User task tech-confirm-defect completed with defectConfirmed=true. State advances to FIELD_VISIT_IN_PROGRESS.

Day 3 — 14:15: Tech removes the old modem. Confirms in app: "source recovered." Worker instance-state-transition fires:

```
POST /api/equipment-instances/inst_01HZ_CM_NJERI_OLD/lifecycle-events
{"toState": "RECOVERED_BY_CONTRACTOR", "toLocationId": "WIK-VAN-contractor_ke_nairobi_a",
 "reasonCode": "CONTRACTOR_RECOVERED_FROM_FIELD", "contractorId": "contractor_ke_nairobi_a",
 "originRefKind": "swap_request", "originRefId": "swap_01HZ_NJERI_MODEM",
 "actorUserId": "tech_franchise_a_03", "expectedEventSequence": 5}
→ 201 {"eventId": "evt_01HZ_NJERI_RECOV", "eventSequence": 6}
```

Worker contractor-stock-route then fires — this is THE Wananchi routing fix:

```
POST /api/stock-movements
{"operatorCode": "WIK", "skuId": "WIK-CM-CISCO-DPC3848", "fromLocationId": null,
 "toLocationId": "WIK-VAN-contractor_ke_nairobi_a", "qty": 1,
 "reasonCode": "RECOVERED_FROM_FIELD", "reference": {"kind": "swap_request", "id": "swap_01HZ_NJERI_MODEM"}}
→ 201 {"stockMovementId": "smv_01HZ_NJERI_RECOV"}
```

The defective modem is now in Franchise A's van (NOT main warehouse). Kafka emits `SwapRequestSourceRecovered`.

Day 3 — 14:20: Tech installs the new modem. Worker advances target:

```
POST /api/equipment-instances/inst_01HZ_CM_NJERI_NEW/lifecycle-events
{"toState":"IN_FIELD_ACTIVE","toAccountId":"acct_01HZ_NJERI",
 "reasonCode":"WO_INSTALL_FINALIZED","contractorId":"contractor_ke_nairobi_a",
 "originRefKind":"swap_request","originRefId":"swap_01HZ_NJERI_MODEM",
 "actorUserId":"tech_franchise_a_03","expectedEventSequence":3}
→ 201 {"eventId":"evt_01HZ_NJERI_PLACE","eventSequence":4}
```

State `TARGET_PLACED`. Worker serial-unbind-broadhub-stub fires:

```
POST /broadhub-adapter/v1/serial/unbind
{"cmMac":"00:11:22:33:44:55","mtaMac":"00:11:22:33:44:57","cpMac":null,
 "slotPort":{"cmts":"nrb-cmts-01","slot":3,"port":12}}
→ 200 {"unbindOk":true,"adapterResponseCode":"OK"}
```

Worker serial-rebind-broadhub-stub fires (UC-PR4: SAME slot/port):

```
POST /broadhub-adapter/v1/serial/rebind
{"cmMac":"00:AA:BB:CC:DD:EE","mtaMac":"00:AA:BB:CC:DD:F0","cpMac":null,
 "slotPort":{"cmts":"nrb-cmts-01","slot":3,"port":12},"doubleConfig":false}
→ 200 {"rebindOk":true,"adapterResponseCode":"OK"}
```

State `PROVISIONING_IN_PROGRESS` → `COMPLETED`. Tech finalizes WO; WO-01 emits `WorkOrderFinalized`; BPMN catches wo-finalized with outcome=`SUCCESS`.

Day 3 — 15:10: Swap-request closes. Kafka emits `SwapRequestCompleted`. NOT-01 sends Njeri: `"Hi Njeri, your modem swap is complete. Your new modem is online. Thanks for your patience."` Franchise A's van now carries the defective unit `inst_01HZ_CM_NJERI_OLD` (state `RECOVERED_BY_CONTRACTOR`), available for batch return-to-main-warehouse processing on Franchise A's next inventory cycle.

8. Workshop sources

- **Workshop 2026-04-28 (Billing deep dive)** — Wallet funding gate for contractor WOs: "tech is dispatched only after the customer wallet is topped up." Drives the pay-first sequence in §2 BPMN and the `AWAITING_PAYMENT` state.
- **Workshop 2026-04-29 (Provisioning / Work Order deep dive)** — Hybrid chassis (HYB) mechanics: CM-MAC = chars 4–15 of chassis Equipment ID formatted with colon every 2 chars; MTA-MAC = CM-MAC + 2; CPMAC = optional public-IP carrier for static-IP HFC packages. Drives the `isHybridChassis`, `sourceMtaMac/targetMtaMac`, `sourceCpMac/targetCpMac` flow_payload fields.
- **Workshop 2026-04-30 (Provisioning deep dive)** — UC-PR4 slot/port preservation: HFC modem authorization bound to slot/port; moving to a different slot/port breaks authorization. Drives R-HFC-2, R-HFC-5, R-OSR-RMA-8 (framework). Also: static-IP HFC requires double-config (Broadhub + Clearcable NOMS). Drives R-HFC-6 and `doubleConfig` flag.
- **Workshop 2026-04-30 (Customer Management deep dive — Equipment-routing gap)** — 40+ stale equipment items today route to main warehouse instead of recovering contractor. Drives the contractor-stock-route worker step at Day 3 14:15 in §7.
- **Workshop 2026-04-30 (Provisioning deep dive — Service Termination)** — Equipment dismantle = separate retrieval WO if customer doesn't bring CPE back. Establishes the pattern: field tech recovers the unit; van location is the destination. Applies symmetrically here (defect swap → tech recovers, van location).
- **BP-08 Equipment Swap (RMA) GAP analysis** — Wananchi anchor: HFC swap = field pickup; UC-PR9 (serial unbind/rebind); G-08-1 (workflow) and G-08-2 (contractor stock traceability). Both addressed by this flow.